

SAFE EXTRACTION OF DEVELOPED PILLARS BY OPENCAST METHOD

INTRODUCTION

Initially coal mines in India were small and owned by small private owners. These mines, even if at shallow depth, were mainly worked by Bord and Pillar (B&P) method of mining using either incline or shaft as mode of entry. The method was practiced because it required less investment, less machine and less skilled manpower compared to longwall technology or surface mining. Most importantly unskilled manpower was cheaply available in the country at that time. B&P method consists of two phases – the development operations and depillaring operations. In the development operation the seam/seams are developed in form of panels by driving strike and dip galleries (bords) forming square (or rectangular) pillars. This is followed by depillaring operation in which the pillars formed during the development phase are extracted in conjunction with caving or with stowing. However, under certain conditions the developed pillars could not be depillared resulting into locking coal in form of developed pillars, and presently such locked coal amounts to few million tonnes. There are nearly 3 to 3.5 BT of coal reserves in standing pillars in the country.

Coal production of the country in financial year 2015-16 is about 638 Mt. The share of opencast mines is more than 90%. Opencast mines are shrinking day by day due to exhaustion of virgin coal. Extraction of coal by Underground mining is very uneconomical.

In future there will not be any virgin coal seam for opencast mining and only alternative for coal production will be by open casting of developed pillars of underground mines.

Surface mining of developed coal seam/seams were preferred because of a number of factors - higher production and productivity, low cost of production, higher percentage of recovery, lower specific investment, more safety, no ventilation problems, no roof and support problems, less illumination problems, high degree of mechanization, less manpower requirement, etc.

Problems Associated with Extraction of Developed Pillars by Opencast Mining

Extraction of developed pillars using opencast method is very challenging for mining engineers. A number of problems are associated with the extraction of developed coal seam/seams by surface mining. Some are more general in nature and rests are site specific.

The problems encountered during mining of the said pillars are discussed below:

1. Uncertain Extraction Ratio

In most of the cases, the reserve of coal determined on the basis of available old plan is not correct but exaggerated. This is because the actual sizes of pillars and galleries are not shown on the plans as the galleries were widened and heightened beyond the statutory limits whereas on the plans these are shown on the basis of statutorily allowable

limits. A realistic estimate of the available coal reserve can be made by making a fresh survey of the existing underground workings which is often difficult and costly affair.

2. Deterioration of Coal Quality

Generally when the coal seams are extracted, there is deterioration in quality of coal. During the extraction of the last overburden bench the galleries get filled by broken waste materials that are cleaned later, but it is not possible to clean the galleries totally and as a result the left over waste materials in the galleries get mixed with the coal thereby deteriorating its quality.

3. Accidental fall of HEMM in Galleries

During the working in last overburden bench it is possible that some HEMM (like shovels, dozers, etc) may fall in the galleries causing a lot of problem. Taking out the entrapped machine is very difficult and time consuming. This type of incidences occurs as the parting between the old working and the bench floor is not able to support the machines or the unsupported span of the parting is too high. To have a correct estimation of the parting thickness is also difficult as the gallery height may be increased due to collapse of roof with the passage of time thereby reducing the parting thickness. To some extent the problem may be avoided if no HEMM is allowed on the floor of the last overburden bench or the exact parting thickness is determined by advanced borehole before any heavy machine is allowed to work on the bench.



Fig. 1: Accidental fall of an excavator in old gallery while removing overburden

4. Drilling and Blasting Problems

A very common problem, while drilling in the last overburden bench is the loss of drill hole as it punctures into the galleries. However, these holes may be blasted using the bottom stemming techniques. Bottom stemming of all the holes in the last overburden bench is a necessity as there is always a possibility that drill hole may puncture into a gallery or it will end up very near to a gallery, and if a hole that has end up near the gallery is blasted without bottom stemming it may result into a blown out shot in the old galleries giving rise to a coal dust explosion.

Apart from the above problem, to get a good fragmentation in the last overburden bench is always a problem. Due to the underground workings the rock mass immediately above the developed coal seam is generally fractured as a result of readjustment of the stresses and when such strata are blasted, a lot of boulders are generated and the explosive energy is not utilized properly.

5. Problems of Fire

In case of the coal seams whose incubation period is short, it is possible that between opening up of coal face in the coal bench and extraction of same, the coal catches fire. In such cases the extraction method should be such that minimum coal is exposed at one time and the coal extraction operation process has to be faster. In addition to this appropriate fire quenching techniques should also be adopted. In some cases fire is already there in the old workings. These fires are to be tackled as the mining progresses.

The problem of fire may also be in the spoil heap due to the presence of coaly material in the waste excavated from the last overburden bench and this requires additional care and incurs extra cost to quench such fires.

Another problem of fire in the coal seams, either because it is already there or because of spontaneous heating, is the problem of hot hole blasting. In such situations all the necessary precautions as stipulated by DGMS are to be taken during the blasting of these hot holes.

- DGMS Circulars issued time to time regarding extraction of developed pillars

1. Precaution against danger of coal dust explosion while extracting pillars by opencast method (Cir. Tech. 4/1980)-Attention is invited to DGMS (Tech) Circular No. 11/1979 in which it was stipulated that LOX shall not be used for blasting in overburden within 15m of the underground workings in the coal seams immediately below or in the vicinity. It was also suggested that:-

- No shot hole shall be drilled in O/B above underground galleries.
- A parting of at least 3m between the bottom of a shot hole and the coal seam shall be left intact.

The above precautions have to be observed not only with LOX but With all other types of explosives used in opencast mines.

After a coal dust explosion in an opencast mine, the previous guidelines were elaborated to prevent a recurrence of the accident. The Precautions to be taken in this connection are listed below-

- *Survey, Cleaning and dusting:* Where the U/G workings are accessible, the same shall be surveyed and cleaned of all coal dust and thickly stone dusted before commencement of extraction of pillars.
- *Withdrawal :* All persons from U/G in the same workings seam or any other seam or section connected therewith shall be withdrawn before commencement of blasting operations in the quarry and the work persons shall not be permitted into the said U/G workings unless and until the same have been inspected by a competent person

OR

The U/G workings to be quarried shall be isolated by explosion proof stoppings or by any other approved manner from any active working area either in the same seam or different seam or section as the case may be, so as to prevent transference of danger to the said working areas.

- *Location of holes:* The spacing of holes in the bench lying immediately above the coal seam (referred to hereinafter as last OB bench) shall be so adjusted that the holes do not lie immediately above the galleries as far as possible in order to ensure that the blast holes do not directly fire into the U/G workings.
- *Safe parting:* The depth of the hole in the last OB bench shall be such as to leave at least 3.0 m thickness of OB above the coal seam, and to ensure compliance with this requirement a pilot hole be put for each round of blasting to determine the total thickness of OB over the coal seam.

NOTE:- This condition has been modified in subsequent DGMS Circular, which is as follows

- No person or machinery shall be deployed or permitted to be deployed in any portion of the quarry lying over U/G galleries, free from water, where the thickness of parting is less than 6 m in coal or 4 m in O/B.
 - To comply this conditions parting of at least 4 m thick immediately above the developed pillars shall be blasted at once including thickness of coal seam to prevent any sinking of HEMM and persons into any U/G galleries during extraction of coal.
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- *Filling sand in holes:* Where there is any doubt and particularly where there is any cracks and crevices, the bottom 2m length of the hole shall be filled with sand.
 - *Delay detonators not to be used:* Unless otherwise permitted by DGMS, no delay detonators shall be used in coal and the manner of extraction of pillars shall be by drilling & blasting holes in coal pillars only from top downwards.
 - *Workings developed in more than one section :* Where more than one section of the seam had been developed on pillars, the shot holes shall not be drilled 3m of a lower section, and care shall be taken that, as far as possible the blast hole do not directly fire into any U/G gallery. To ensure compliance with requirement, a pilot hole shall be put for each round of blasting.
 - *Use of water ampoules/moist sand:* All holes in the last OB bench and/or in coal shall be charged with water ampoules or with moist sand of at least 0.6m in length at the bottom of the hole.
 - *Shelter:* No person including shot firer take shelter within 100m of the quarry openings, and such shelter shall be of an approved design.
 - *Administrative control:* The opencast working over U/G part of working shall be under the administrative and technical direction and control of the Manager and Agent.

2. Blasting in Fire areas (Cir.Tech.2/1985 & 2/1990): A number of fires are known to be active in different Coalfields. Any attempt to win coal from these areas involving blasting is fraught with danger. In a major accident, 6 persons were killed when a NG- based explosives was being used in an opencast coal mine where fire existed. Some trial were carried out by R&D wing of DGMS in collaboration with CMPDIL and CMRS in extraction of coal from fiery opencast mine and recommended the following precautionary measures:-

- No explosives other than slurry and emulsion explosives shall be used.

Blasting shall be done with detonating fuse down the hole.

- Temperature inside the blast holes shall be measured (before filling with water) and, if the temperature exceeds 800c, in hole, such hole shall not be charged. Records of measurements of temp. in each hole shall be maintained in a bound paged book.
- All blast holes shall be kept filled with water. When any hole is traversed by cracks or fissures, such hole shall not be charged unless it is lined with asbestos pipe and the hole filled with water. In addition, bentonite should be used for sealing any cracks at the bottom of the holes.
- Detonating fuse shall not be laid on hot ground without taking suitable precautions, which will prevent it from coming in contact with hot strata.
- The charging and firing of the holes in any one round shall be completed expeditiously and in any case within 2 hours.
- Blasting operations shall be carried out under the direct supervision of an Assistant manager.

3. Firing of shots in fire areas/in vicinity of hot strata in opencast Coal mines [DGMS (Tech) Circular 4/2006]:-

The following additional precautionary measures shall be taken over and above the measures stipulated vide DGMS (Tech.) circular 2/1985.

- The blasting officer shall ensure accurate measurement of temperature at bottom of the shot holes drilled in coal benches and in O/B benches immediately above the coal seam and other fiery areas to determine whether fire exists. The measurement shall be recorded in a bound paged book kept for the purpose. For this purpose the temperature shall be measured initially after completion of the hole and thereafter at least once every shift. No hole shall be charged, if the temp. in the hole exceeds 800C.
- As a precaution against presence of pocket fires in coal benches in developed seams, O/B benches immediately above the coal seams and other fiery areas in the mine, the explosives charged shall be fired by detonator attached to the detonating cord at the surface and not within the shot hole.
- All explosives, cast boosters, detonators, detonating cord and shock tubes shall be subjected to proper testing in an approved laboratory in respect of temperature sensitivity and impact sensitivity for safe handling in mines. A certification to that effect shall be supplied for each batch.
- Sleeping of holes shall not be permitted in coal benches, where fire or Spontaneous heating has been detected and in O/B benches lying immediately above such seam and at all places where there is likelihood of fire or Sp. Heating.

- No PETN/TNT based cast booster shall be used for initiating non- cap sensitive slurry/emulsion explosives in coal benches and O/B benches of a fiery coal seam.

- **METHODS FOR EXTRACTION OF DEVELOPED COAL SEAMS**

A coal seam, depending on its thickness and roof and floor conditions may be developed in one of the following modes or a combination of these:

1. The seam is developed for its full height. (fig. 3a)
2. The seam is developed leaving roof coal. (fig. 3b)
3. The seam is developed leaving roof coal. (fig. 3c)
4. The seam is developed in the middle. (fig. 3d)
5. The seam is developed in multi section. (fig. 3e)

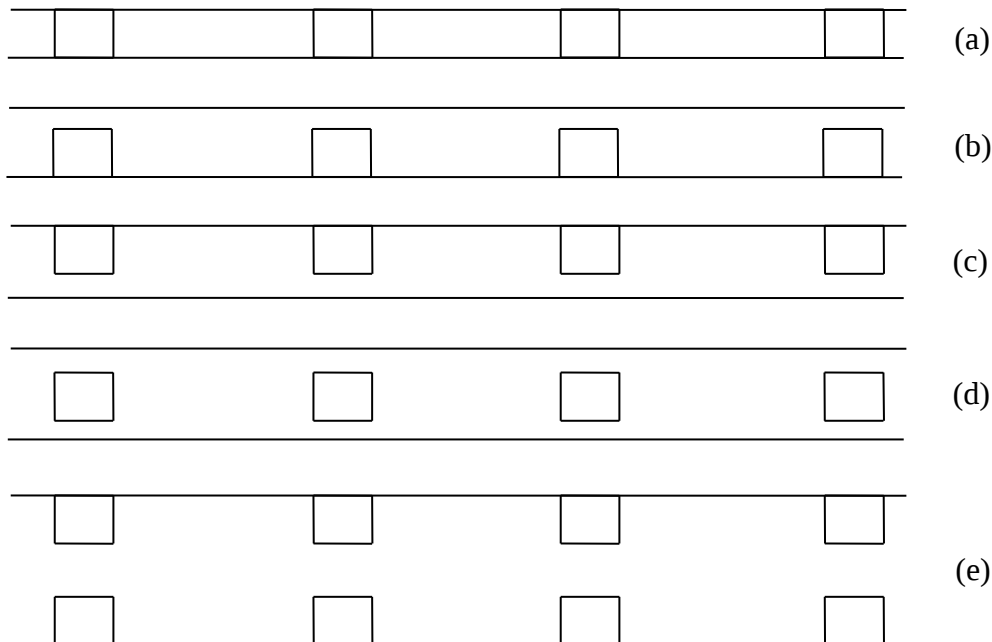


Fig. 3 Different Modes of Developing a Coal Seam in B&P

Different zone of working

For safe and systematic excavation, the mine is divided into four zones viz.

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|---------------------|--------|
| A. SAFE ZONE. | Method |
| B. CRITICAL ZONE | |
| C. DANGER ZONE AND; | |
| D. COMPACTED ZONE | |

(A) SAFE ZONE: It is the zone of safety, where the parting is over U/G galleries is more than 15m. In this zone all activities can be carried out and the zone is to be demarcated with `White` flags.

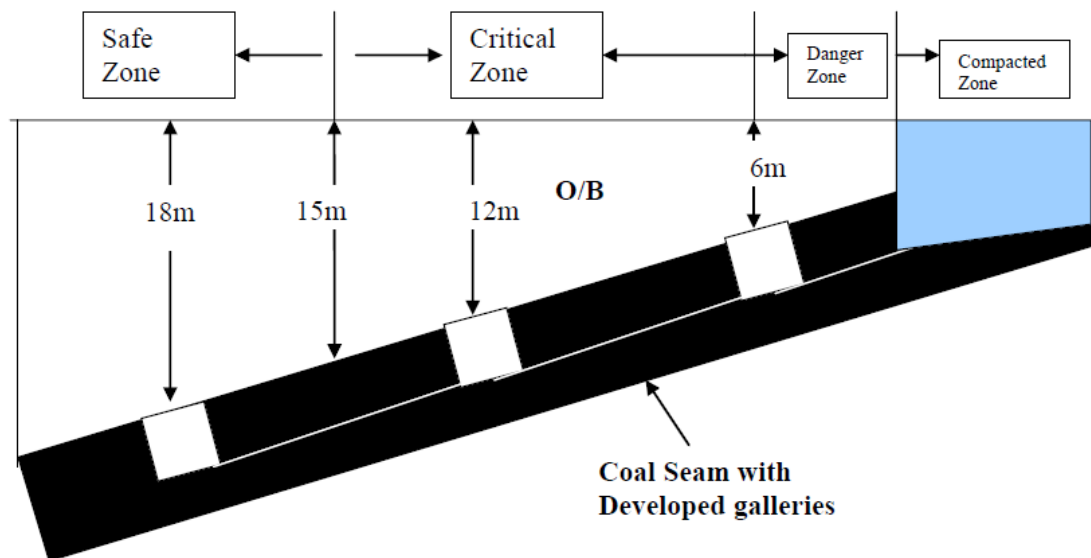
(B) CRITICAL ZONE: It is the zone, where proper precautions are required for excavation. In this zone, the parting of working O/B benches over U/G galleries is more than 4m and less than 15m. Whenever, the plan parting reaches 15m to u/g galleries, further reduction shall not

be done unless it is proved physically. After proving the physical parting, further parting is reduced to 4m hard O/B. In this zone the excavation is done by Back hoe shovel in level loading. This zone is demarcated with `Yellow` flags.

(C) DANGER ZONE: Danger zone means, the area where the parting over U/G galleries is 4m hard O/B or less. No work shall be done in this zone except dozing and drilling for compaction. The dozing and drilling shall be done under proper supervision. Danger zone is to be demarcated with `Red` flags.

(D) COMPACTED ZONE: When danger zone is converted to safe zone by Compacting the U/G galleries by drilling and blasting (4m O/B and coal seam combinedly), it is termed as `Compacted` zone. This zone is permitted for normal extraction of coal/OB, as it is safe. It is demarcated with White flags.

Note: As per DGMS guideline, the extraction of O/B in Critical zone should be done upto 4m parting over galleries, but 4m hard O/B parting is not safe for running HEMM and formation of pot-hole can't be avoided. Hence for extra precaution, Critical zone is assumed between 6-15m. When the working in critical zone reaches to 6m, it is treated as danger zone.



There is no single known method for extraction of developed coal seams by surface mining. Depending on the site conditions, it is necessary to adopt a method of working which in all likelihood will ensure uninterrupted working of HEMM as far as practicable, minimize the dilution of coal exploited and provide security against accidental falling of equipments into galleries.

Methods without Integrated Blasting

In case of seams which have been developed for its full height or developed along the roof either in single section or in multi-section, the following method may be tried.

The overburden may be excavated by conventional benching method using shovel dumper combination except in the last bench just above the developed coal seam. Depending on the conditions prevailing, the last overburden bench should be excavated either by dragline (Dip < 1 in 20) or by backhoe operation.

Blasting in the last overburden bench should be done in such a manner that the galleries of the developed seam are filled with blasted rock. In this bench, as far as practicable, the drilling has to be done carefully so that the drill holes do not get open in the galleries or end

very near to a gallery. However, if a hole opens up in a gallery the hole may be blasted using bottom plugging techniques and all the holes should be bottom stemmed.

After the blasting is done, the muck pile is excavated by either dragline or backhoe (depending on the prevailing conditions) upto the roof of the coal seam.

In the next step the drilling of the exposed coal pillars is completed and thereafter the filled galleries are cleaned by backhoe. Then the standing pillars are blasted and coal is handled with the help of shovel dumper or pay-loader dumper combination.

The different steps of this method are explained in figures 4, 5 and 6.

In case of the developed seams where coal is left in the roof, the extraction method will be very much similar to the method described earlier, but in this case the galleries will not be filled by the blasted overburden material. After the blasting of the last overburden bench the muck pile is excavated by dragline or backhoe (as the case may be) upto the roof of the coal seam.

In the next step when the drilling has to be done in the coal seam, care has to be taken in respect of movement and positioning of the drills so that it does not fall in the old galleries and the holes are drilled in the pillars. After blasting in the coal bench the coal is handled with the help of shovel dumper or pay-loader dumper combination.

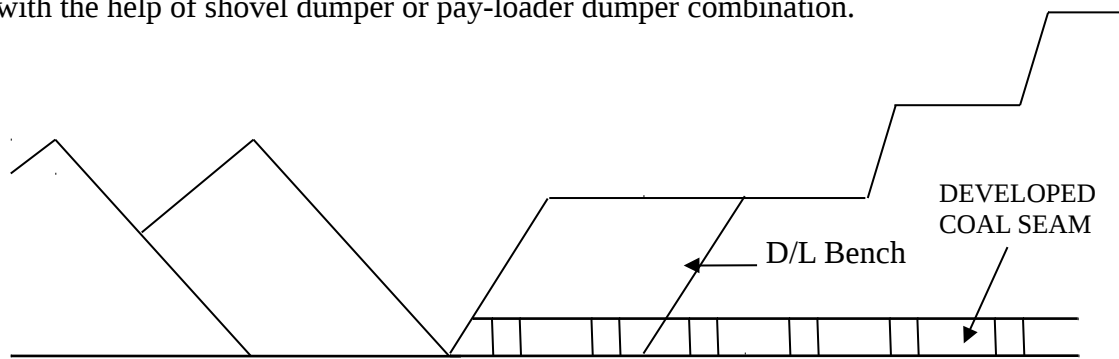


Fig. 4 Excavation of last Overburden Bench by Dragline

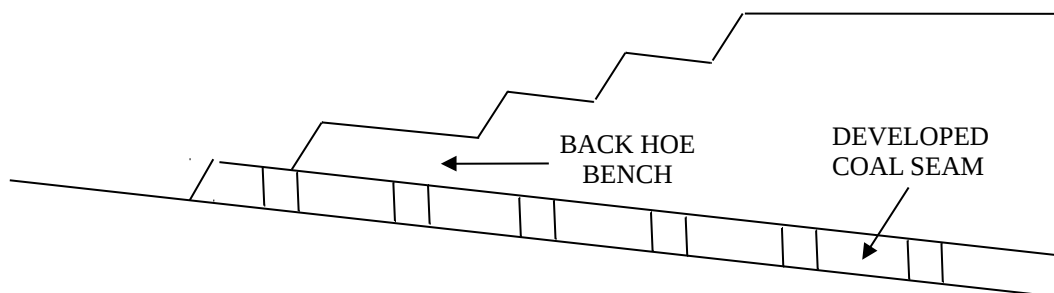


Fig. 5 Excavation of last Overburden bench by Backhoe

Fig. 6 Plan and Section of Working by Dragline in the last Overburden Bench

